

UNIVERSITY OF PROFESSIONAL STUDIES, ACCRA (UPSA)

DEGREE PROGRAMME

University of Professional Studies, ~~Accra~~ **REGULAR SCHOOL**

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END OF SECOND SEMESTER EXAMINATIONS- 2016/2017 ACADEMIC YEAR

FACULTY OF ACCOUNTING & FINANCE

DEPARTMENT OF BANKING & FINANCE

LEVEL 200

BCPC 208: QUANTITATIVE METHODS

Date: Monday, May 29, 2017.

TIME ALLOWED: 3 HOURS

INDEX NUMBER

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INSTRUCTION(S)

There are Two (2) Sections in this paper, Sections A & B.

Answer All Questions in Section A and Any Three (3) from Section B

SECTION A

ANSWER ALL QUESTIONS

QUESTION ONE

- a) A company that produces mobile phones has its demand and supply functions as $P = 28 - 2Q_d$ and $P = 4 + 4Q_s$.
- (i) Determine the equilibrium price and quantity for mobile phones. (2 marks)
 - (ii) Suppose a fixed tax of ₵6 per unit is imposed on the good. Determine the new equilibrium price and quantity. (2 marks)
 - (iii) How do the producers and consumers share the tax? (1 mark)
- b) Find the value, in 5 years' time, of \$10 500 invested at 8% interest compounded after every three months. (3 marks)
- c) A company's monthly sales figures for January are ₵6800. This is expected to fall for the following 9 months at a rate of 3% each month. Find the amount of sales for the month of June. (3 marks)
- d) The production function for a firm is given as $Q = 4LK - 3L^2 - 2K^2 + 6L + 14K$ where Q is the total output, K is capital and L is labour. Find an expression for marginal product of capital and marginal product of labour. (4 marks)

SECTION B

ANSWER ANY THREE (3) QUESTIONS

QUESTION TWO

A young entrepreneur who is a graduate from one of the universities in Ghana uses three ingredients: pepper, ginger and cloves to produce three brands of spices namely Sweetpep, Gingoep and Clovipec in satchets to supply some Waakye Sellers in Accra. In producing one satchet of Sweetpep, the enterprise will require 10g each of pepper, ginger and cloves. Similarly, producing Gingoep requires 30g each of pepper and cloves together with 40g of ginger while producing Clovipec requires 30g each of pepper and ginger, and 40g of cloves. In a particular day, the company has a total of 35000g, 32000g and 37000g of cloves, pepper and ginger available respectively. (*Hint: Summarize the imports of the question into a table*)

- a) Write three linear equations and hence form a matrix to represent the above problem.

- b) Find the determinant of the matrix.
- c) Determine the number of bags of each brand of spices that should be produced from the available raw materials using the Cramer's rule or matrix inversion.

QUESTION THREE

Exquisite Furniture Company produces chairs and tables for two outlets in Ghana. It takes 3 hours and 6 hours of labour time to produce a chair and a table respectively. Finishing these products requires 2 hours of finishing time for a table and 4 hours for a chair. Besides, it takes 2 hours to thoroughly polish a chair and check to ensure it meets required standards and the same number of hours for doing same for a table. It costs the company ₵120 and ₵80 to produce a table and a chair respectively. The market conditions dictate the selling price of a chair and a table at ₵130 and ₵190 respectively. Typically in a month, the company has 120, 270 and 200 hours of polishing time, labour time and finishing time available respectively. The company aims to attain the greatest possible profit from the production of tables and chairs in a week. *(Hint: Summarize the imports of the question into a table)*

- a) Formulate a linear programming model for the above managerial problem.
- b) Using the simplex method or otherwise, find how many tables and chairs the company should produce and the largest profit this production.
- c) Determine the shadow price of the labour time available to the company.

QUESTION FOUR

A soft drink manufacturing company incurred a fixed cost of ₵2 before its first bottle of ginger drink. The marginal cost function of the firm is $MC = 3Q^2 - 16Q + 20$ and its marginal revenue function given as $MR = 20 - 4Q$ where Q is the level of output produced. All figures associated with the firm are measured in ten thousands of Ghana Cedis. Find

- a) The firm's total cost function.
- b) The output level at which the firm minimizes its total cost and the minimum cost.
- c) The output level and the price at which the firm maximizes profit, and its maximum profit.

QUESTION FIVE

A monopolistic producer is contracted to produce and sell a total of 1005 textbooks of either Intermediate Science or Mathematics textbooks to two educational institutions namely, Royal

Academy and St. Peter's High School, at different prices. The joint total cost function of production is given by $TC = Q_1Q_2 + 10Q_1 + 10Q_2$ where Q_1 and Q_2 are quantities of Science and Mathematics textbooks produced respectively. If the price at which the producer sells to Royal Academy is given as $P_1 = 90 - Q_1 + Q_2$ and the price at which he sells to St. Peter's High School is given as $P_2 = 70 - Q_2 + 2Q_1$ where P_1 and P_2 are the respective prices.

- Find the maximum profit if the firm is contracted to produce a total of 1005 textbooks of either type to the two educational institutions.
- Estimate the new optimal profit if the production quota rises by one extra textbook.

(Assume that second order conditions are satisfied)

QUESTION SIX

Kwarteng, a UPSA student took a student loan totaling ₵5000 with an interest of 8% per annum. He has put in place a payment plan that continuously deducts money from his account at a rate which amounts to ₵2000 per year.

- Write a first-order linear differential equation to represent the above problem.
- Hence find how long it will take for the loan to be totally paid off.

FORMULA SHEET

$$a_n = a + (n-1)d; \quad S_n = \frac{n}{2}[2a + (n-1)d]; \quad a_n = ar^{n-1}; \quad S_n = \frac{a(1-r^n)}{1-r}$$

$$A_t = P\left(1 + \frac{r}{m}\right)^{mt}; \quad A_t = Pe^{rt}; \quad i = \left(1 + \frac{r}{m}\right)^m - 1; \quad i = e^r - 1$$

$$\text{For } \frac{dy}{dt} + vy = Z \Rightarrow y(t) = e^{-\int v dt} \left(A + \int Z e^{\int v dt} dt \right)$$

End of Paper

Examiners: Mr. Mawunyo Agradi, Mr. Joseph Adu, Dr Ebo Botchway, Dr Victoria N. Sam, Mr. Kofi Afriyie Nyamekye, Mr. Alhassan Bunyaminu and Mr. Richard Agbanyo